SCX/EGP MoE

2026-06-25

2023-2026 Mixture-of-Experts (MoE)Knowledge Distillation (KD)MLIP gauge Expert Governance

SCX/EGP SCX/EGP

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1. A. Mixture of Experts (MoE) in MLIP
2. B. Knowledge Distillation for MLIP
3. C. Expert Merging / Gauge / Energy Alignment
4. D. Expert Compilation / GovernanceSCX
5. E.

A. Mixture of Experts (MoE) in MLIP

A1. DeepMD MoE/MoLE — "Scaling MLIP with Mixtures of Experts"

Liu, Zhang, Peng, E, Zhang, Wang. "Scaling Machine Learning Interatomic Potentials with Mixtures of Experts." arXiv:2603.07977 (2026).

MoE MLIP MoE NLP CV MoE

- **MoLEMixture-of-Linear-Experts MoE MLP** - element-wise routing vs. configuration-level routing vs. MoE - **shared expert** - DeepMD-kit DP

- **E(3) equivariant routing** MoE - MoE + MoLE - - OMol25OMat24OC20M SOTA

SCX/EGP - DeepMD MoE SCX/EGP - MoE DeepMD MoE MoE SCX/EGP —— - DeepMD MoE emergent expertise SCX/EGP pre-defined expertise

SCX - SCX/EGP

DeepMD MoE - EGP MACE + NEP + SevenNetDeepMD MoE DP - EGP DeepMD MoE

- DeepMD MoE ——SCX/EGP ”” - DeepMD MoE ——SCX/EGP MoE ”MoE + ”

A2. Allegro/NequIP MoE — Spatial Partition + Co-training Agreement

Nascimento et al. “Mixture of Experts Framework in Machine Learning Interatomic Potentials for Atomistic Simulations.” arXiv:2604.26143 (2026).

MoE artificial stress fields

- E(3)-equivariant **Allegro** MoE - **Spatial Partitioning**

- **Co-training agreement constraints**—— - Pt+CO

- exact energy conservation maintained - EOS - - **2**

SCX/EGP - SCX/EGP - - ”” SCX/EGP ”Gauge ”——

SCX - SCX/EGP MoE - SCX/EGP

- ””SCX/EGP - ——

A3. MoE — Shazeer et al. 2017, Switch Transformer

- Shazeer et al. “Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer.” ICLR 2017. - Fedus, Zoph & Shazeer. “Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity.” JMLR 2022.

Shazeer 2017 **MoE** top-k ””Switch Transformer top-1

- **Noisy Top-K Gating** logits top-k - **Switch Transformer** ”” - **Capacity Factor** token - router float32 dropout

Mixtral 8x7BDeepSeek-V2/V3Llama 4 MoE LLM

SCX/EGP - MoE SCX/EGP ”” - NLP MoE token - SCX/EGP - NLP MoE FFN SCX/EGP

SCX - SCX/EGP ——**“”**—— MoE 1. MoE collapse 2. - SCX/EGP

- MoE SCX/EGP - NLP MoE ””shared expert SCX/EGP

B. Knowledge Distillation for MLIP

B1. MACE → NEP

“Constructing machine learning interatomic potentials with minimum amount of ab initio data.” npj Computational Materials, s41524-026-02023-y (2026).

DFT ~200 MLIP DFT

- **-** - Stage 1 **MACE-MP-0** ~200 DFT - Stage 2 MACE **NEP** - LGPSLAT-PLYC

single-shotNEP MACE MD

SCX/EGP - ” → ”SCX/EGP ” → ” SCX/EGP 1. **Gauge** SCX/EGP Gauge 2. SCX/EGP →→

SCX - SCX/EGP ”” - SCX/EGP MACENequIPSevenNet

B2. Ensemble Knowledge Distillation (EKD) for MLIP

Matin et al. “Ensemble Knowledge Distillation for Machine Learning Interatomic Potentials.” arXiv:2503.14293 (2025).

QC

1. QC **MLIP**/ 2. ensemble-averaged forces 3. **QC +** 4. ANI-1ccx COMP6 SOTA

SCX/EGP - ””SCX/EGP ””EKD

SCX - EKD SCX/EGP - EKD SCX/EGP ””

B3. Foundation Model Distillation

Gardner et al. "Distillation of atomistic foundation models across architectures and chemical domains." arXiv:2506.10956 (2025).

distillation via synthetic data

- large universal MLIP - - >10xGNN→GNN >100xGNN→ACE -

SCX/EGP - "→"SCX/EGP "→"—— SCX/EGP

- "" SCX/EGP - ""—— SCX/EGP

B4. Teacher-Student Training for MLIPs (RSC 2025)

Matin et al. "Teacher-student training improves the accuracy and efficiency of machine learning interatomic potentials." RSC Digital Discovery, 2025.

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- **HIPNN**Hierarchically Interacting Particle Neural Network - - - **>2 <50%**
- Pareto dominance

SCX/EGP - SCX/EGP

- SCX/EGP ""—— **EGP synergy**

B5.

			SCX/EGP
SevenNet-Nano (arXiv:2604.10887)	2026	SevenNet-Omni Nano>10	
PFD (Phys. Rev. Materials, 2025)	2025	Pre-train→Fine-tune→Distill ~100 DFT ~100x	PFD

			SCX/EGP
ARK Distillation (npj Comput. Mater., 2026)	2026	+ 11.9x	

ARK Distillation - Lim et al. “Angular relational knowledge distillation of machine learning interatomic potentials for scalable catalyst exploration.” npj Computational Materials, 2026. - **angular relational vectors** PES - 58 11.6 GPU-hours59.4 CPU-years - SCX/EGP ”” PES

PFD - Wang, Gao et al. “Pre-training, fine-tuning, and distillation (PFD): Automatically generating machine learning force fields from universal models.” Phys. Rev. Materials, 2025. - DPA-2 → → DeePMD ~100 DFT - SCX/EGP PFD PFD ””SCX/EGP ””

C. Expert Merging / Gauge / Energy Alignment

C1. Model Merging (Model Soup, Git Re-Basin,)

Model Soups (Wortsman et al., ICML 2022)	2022	/
Git Re-Basin (Ainsworth et al., ICLR 2023)	2023	permutation alignment basin
ZipIt! (ICLR 2024)	2024	
AME (Lee & Ngo, NeurIPS 2025)	2025	soup ”Amortized Model Ensembling“

Soup-of-Experts (ICML 2025)	2025		
LLM Souping (Meta, 2025)	2025	LLM SoCE	

SCX/EGP - SCX/EGP ”” - weight-space averaging - SCX/EGP prediction-space ensembling - SCX/EGP

SCX - SCX/EGP MACE + NEP + SevenNet - ”” MLIP SCX/EGP - SCX/EGP

- MLIP - ”MLIP ”——

C2. MLIP Gauge Freedom Species Energy Shift

“Gauge dependence of total energies and cross-functional transfer learning.” arXiv:2504.05565 (2025).

MLIP **Gauge gauge freedom** DFT ”Gauge” MLIP

- ” AtomRefspecies energy shift“ - **refit AtomRef GNN** AtomRef Pearson 0.09 0.93 - Method 4refit AtomRef + train GNN MAE = 17 meV/atom GNN 26 meV/atom

SCX/EGP - Gauge freedom SCX/EGP **Gauge** MLIP OUTCAR ”energy without entropy” MACE eV - SCX/EGP Gauge GAU

SCX - AtomRef SCX/EGP Gauge —— - SCX/EGP Gauge DOF

- ”formation energy” Gauge-invariant SCX/EGP formation energy total energy - **Gauge** ——SCX/EGP - GGA/GGA+U → r²SCANSCX/EGP Gauge SPME → MACE → NEP

C3. Species Energy Shift

MLIP

CHGNet	AtomRef + GNN	+
M3GNet	AtomRef + GNN	
NequIP	AtomRef (subtract per-atom energy)	
CACE	AtomRef + GNN	Composible Atomic Cluster Expansion
MACE-MP-0		MP
SevenNet	offset	

——SCX/EGP Gauge

D. Expert Compilation / GovernanceSCX

SCX/EGP

D1. “”Expert Normalization Before Distillation

- MACE→NEP, SevenNet-Nano, PFD, ARK, EKD
- EKD
- **””**
- SCX/EGP Gauge → Route → Compile ””””

SCX - SCX/EGP ””SCX

D2. Expert Domain Certification

MLIP

- AI ”AI ”AIGP / MLIP
- “What Makes An Expert?” (arXiv:2411.00179, 2024) ML ””
- **”MLIP ”**
- SCX/EGP ECR
  SCX

SCX - ”MLIP Expert Domain Certification” - conformal prediction

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D3. Multi-Expert Conflict Arbitration

ML MLIP

ML

King Solomon Protocol (2026)		
HCDF (2026)		→→
Cognitive Hives (2026)		//
LLMediator / Mediator (2025)	LLM	+ MoE
AEGIS-Nexus (2026)		
Multi-Expert Fusion Networks survey		top-k

MLIP

- ”” ML-MIX, MoE””
- SCX/EGP Route RTE——

SCX - ML HCDF LLMediator MLIP - EGP

D4.

- MoE ””
- Active learning ALIP ””
- DeepMD MoE ””——
- **””expert performance profile over input conditions**

SCX - ECR -

E.

E1.

	DeepMD MoE (2603)	Allegro MoE (2604)		MLIP	SCX/EGP
MoE	Y	Y	N	N	N
/	N	N	Y	Y	**Y**
	N	N	N	N	**Y**
Gauge	N	Y()	N	N	**Y**
	Y()	Y()	N	N	**Y**
	N	N	N	N	**Y**
	N	N	N	N	**Y**
	N	N	N	N	**Y**
	N	N	Y	N	**Y**

SCX/EGP ”” SCX/EGP SCX/EGP

E2.

	DeepMD MoE (2603.07977)	DeepMD
	Allegro MoE (2604.26143)	”””” EGP
	ARK (npj 2026)	ARK ””
	PFD (PRM 2025)	PFD ”” EGP
		Compile
	(ML)	
	Foundation (2506.10956)	→

E3. SCX/EGP

1. **Prerequisite Normalization**
   - SCX/EGP
   -
2. **Expert Domain Certification**
   - ””

    - conformal prediction
    - VASP
3. **Cross-Architecture Compilation**
    - SCX/EGP MACE + NEP + SevenNet + NequIP
    - ——DeepMD MoE
4. **Synergistic Distillation**
    - EGP
    - ”EGP > > ”
5. **Input-Conditioned Benchmark**
    - ””
    - NEB
    - ””

E4.

P0	SCX	arXiv SCX/EGP ”Expert Compilation”
P0	EGP	VASP + MACE + NEP Gauge→Route→Compile
P1	Gauge	MLIP Gauge freedom
P1	vs. DeepMD MoE	OMat24 DeepMD MoE
P2		Expert Domain Certification
P2		EGP Route

E5. /

		SCX
DP Technology / AIS (,)	DeepMD MoE (2603.07977)	—— MLIP
Boris Kozinsky (Harvard)	Allegro/NequIP, MoE (2604.26143)	——
InstaDeep	mlip JAX (2605.22698)	——
LANL (Sakib Matin)	EKD, Teacher-student RSC	——

		SCX
Volker Deringer (Oxford)	Foundation distillation (2506.10956)	——
Seoul National Univ. (Jeong Woo Han)	ARK Distillation	——
USTC ()	PFD (2502.20809)	——PFD ”” EGP

A. MoE in MLIP

1. Liu, Y. et al. “Scaling Machine Learning Interatomic Potentials with Mixtures of Experts.” arXiv:2603.07977 (2026). https://arxiv.org/abs/2603.07977
2. Nascimento, G.M. et al. “Mixture of Experts Framework in Machine Learning Interatomic Potentials for Atomistic Simulations.” arXiv:2604.26143 (2026). https://arxiv.org/abs/2604.26143
3. Shazeer, N. et al. “Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer.” ICLR 2017. https://arxiv.org/abs/1701.06538
4. Fedus, W., Zoph, B. & Shazeer, N. “Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity.” JMLR 2022. https://arxiv.org/abs/2101.03961
5. Birks, T., Swinburne, T. & Kermode, J. “Efficient and accurate spatial mixing of machine learned interatomic potentials.” npj Computational Materials (2025).
6. Nascimento, G.M. & Descoteaux, B. “Mixture-of-Experts Framework in MLIP for Atomistic Simulations.” (2026).

B. Knowledge Distillation for MLIP

7. “Constructing machine learning interatomic potentials with minimum amount of ab initio data.” npj Computational Materials, s41524-026-02023-y (2026). https://www.nature.com/articles/s41524-026-02023-y

8. Matin, S. et al. “Ensemble Knowledge Distillation for Machine Learning Interatomic Potentials.” arXiv:2503.14293 (2025). https://arxiv.org/abs/2503.14293
9. Gardner, J.L.A. et al. “Distillation of atomistic foundation models across architectures and chemical domains.” arXiv:2506.10956 (2025). https://arxiv.org/abs/2506.10956
10. Matin, S. et al. “Teacher-student training improves the accuracy and efficiency of machine learning interatomic potentials.” RSC Digital Discovery (2025). https://pubs.rsc.org/en/content/articlehtml/2025/dd/d5dd00085h
11. Kim, J. et al. “A Lightweight Universal Machine-Learning Interatomic Potential via Knowledge Distillation for Scalable Atomistic Simulations (SevenNet-Nano).” arXiv:2604.10887 (2026). https://arxiv.org/abs/2604.10887
12. Wang, R., Gao, Y. et al. “Pre-training, fine-tuning, and distillation (PFD): Automatically generating machine learning force fields from universal models.” Phys. Rev. Materials 9, 113802 (2025). https://arxiv.org/abs/2502.20809
13. Lim, H. et al. “Angular relational knowledge distillation of machine learning interatomic potentials for scalable catalyst exploration.” npj Computational Materials 12, 1 (2026). https://link.springer.com/article/10.1038/s41524-026-02062-5
14. “Data-Efficient and Fast Machine Learning Molecular Dynamics through Integrated Active Learning and Knowledge Distillation.” ChemRxiv (2026).

C. Expert Merging / Gauge Alignment

15. “Gauge dependence of total energies and cross-functional transfer learning.” arXiv:2504.05565 (2025). https://arxiv.org/abs/2504.05565
16. Wortsman, M. et al. “Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time.” ICML 2022.
17. Ainsworth, S. et al. “Git Re-Basin: Merging Models modulo Permutation Symmetries.” ICLR 2023.
18. Lee, S.H. & Ngo, R. “On Defining Neural Averaging.” NeurIPS 2025. https://arxiv.org/abs/2508.14832
19. Ablin, P. et al. “Soup-of-Experts.” ICML 2025.
20. Franke et al. “Hierarchical Re-Basin.” ESANN 2024.

D. Governance / Arbitration (General ML)

21. Grassini Grimaldi, A. "The King Solomon Protocol: A Framework for Human Supervised Arbitration in Multi-Model AI Systems." (2026).
22. Bie et al. "Knowledge-guided policy arbitration: A hierarchical cognitive framework for safety-critical decision-making." Expert Systems with Applications (2026).
23. Hettiarachchi. "From Monolithic Models to Cognitive Hives: A Framework for Multimodal Reasoning Systems." JCSTS (2026).
24. "Beyond Monoliths: Expert Orchestration for More Capable, Democratic, and Safe Large Language Models." arXiv:2506.00051 (2025).
25. "Toward Calibrated Mixture-of-Experts Under Distribution Shift." arXiv:2606.20544 (2026).

2026-06-25 arXivNature PortfolioRSCAPSSpringer SCX/EGP ""
DeepMD Kozinsky